When the diaphragm is stressed,the more amount of petrol fuel from the tank is injected to the engine through the venturi. When the driver released the accelerator pedal, diaphragm get relived from the stress and comes to its original position. Because of this a vacuum created inside the pump chamber will sucks petrol fuel and sent to the float chamber with help of pump valve. By this way pump get actuated automatically for the next process. Fig 9.5.5(e) shows the Acceleration Circuit

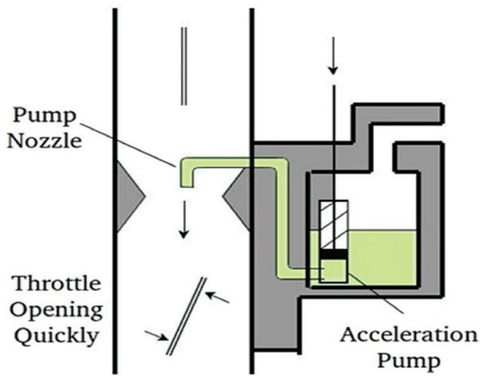



**Fig.9.5.5(e) Acceleration Circuit**

## 9.6 PETROL INJECTION

There are some disadvantages faced by sending air fuel mixture with carburation technique in petrol engines. For example, in multi cylinder engines the amount of air fuel mixture was varied from cylinder to cylinder. So at the slow speed engine could not attain steady state condition. Furthermore the energy produced from the each and every cylinder is getting varied. Hence engine vibrates more. So now days in petrol engines also direct injection of petrol is practiced.

Robert Bosch is a German engineer who introduced first time the concept of direct injection of petrol in the Mercedes racing cars. Comparably Petrol injection technique is completely varied from diesel injection. Petrol injection was made near to the intake manifold with minimum pressure in the injector whereas diesel injector operates with high pressure for injection.

### 9.6.1 Advantages of Petrol Injection technique

1. We can inject uniformly the air-fuel mixture with required amount to the each cylinder in the multi cylinder engine.
2. Increases in volumetric efficiency.
3. It reduces air-fuel mixture escapes from the exhaust during scavenging process.
4. Reduces knocking.
5. For all the speed and torque conditions, uniform amount of air fuel mixture can be supplied.

**The following four parts are mainly used in petrol injection technique:**

1) Pumping element – used for pumping petrol from tank
2) Metering element – it measures the petrol fuel quantity
3) Mixing element – it atomize the petrol fuel and tend to fine mix with air.
4) Distributing element – supplies petrol fuel to all the cylinders uniformly.

Electronic fuel injection system was established for the proper functioning of the engine. In this technique mechanical injector was replaced with electronic fuel injector with control valve.

The layout of electronic fuel injection system was shown in the fig. 9.6.



AM_Unit_09.indd 185 31-01-2019 18:02:29